Retail Sales Intelligence Platform

Business Problem Definition | Global Superstore | 2011–2014

THE BUSINESS PROBLEM

Retail management needs a consistent, evidence-based view of how sales translate into profit across products, categories, customers, regions and time. Aggregate revenue and profit totals alone do not reveal which areas drive performance, where losses occur, or which operational and commercial factors may be associated with weak profitability.

Decision to support

Determine where the business is generating sales and profit, where performance needs closer investigation, and which product, customer, regional or pricing areas should receive management attention. The analysis must distinguish observed patterns from hypotheses; correlation alone will not establish root cause.

PROJECT OBJECTIVE

- Build a reproducible view of sales, profit, order volume and average order value.
- Compare performance by product/category, customer segment, region and time.
- Identify loss-making products and examine discount and shipping-cost patterns.
- Translate verified findings into prioritized, measurable business actions.

KEY BUSINESS QUESTIONS

01 How do sales and profit change year by year and month by month?
02 Which categories, sub-categories and products drive sales and profit?
03 How do regional results differ in revenue and profitability?
04 Which customer segments and customers contribute most to sales and profit?
05 Which products are consistently loss-making despite their sales volume?
06 How are discounts and shipping costs associated with profit outcomes?

Analytical scope and deliverables

- Measures: total sales, total profit, distinct orders and customers, average order value, discounts, shipping cost and product counts.
- Dimensions: order date, region, category, sub-category, product, customer and segment, using the columns present in the supplied dataset.
- Methods: data-quality validation, SQL aggregation, Python exploratory analysis, and Power BI/Tableau visual reporting where source files support them.

VALIDATION GATE Supplied CSV files and SQL output screenshots show differing totals. KPI values must be reconciled against one confirmed source before final findings are published. This brief defines the business question; it does not claim that any suspected cause has been proven.

Expected outcome: a trusted performance baseline and a prioritized set of evidence-linked opportunities for investigation—not unsupported promises of cost savings or revenue growth.